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Section: BSIT22M3

LAB EXAM MIDTERMS SET A DENORMALIZED TABLE

Server: 127.0.0.1 > Database: labexammidterm > Table: sales_record

Showing rows 0 - 3 (4 total, Query took 0.0020 seconds)

SELECT * FROM `sales\_record`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

OrderID	OrderDate	CustomerID	CustomerName	CustomerAddress	CustomerPhone	ProductID	ProductName	ProductCategories	UnitPrice	Quantities	SupplierName
1001	2025-01-10	C001	Ana Cruz	Cavite City	09171234567	P01, P02	Keyboard, Mouse	Accessories, Accessories	800, 500	1, 2	TechSource, TechSource
1002	2025-01-12	C002	Juan Dela Rosa	Imus Cavite	09981234567	P03	Monitor	Display	7500	1	ScreenPro
1003	2025-01-12	C001	Ana Cruz	Cavite City	09171234567	P02, P04	Mouse, USB Drive	Accessories, Storage	500, 600	1, 3	TechSource, DataHub
1004	2025-01-15	C003	Peter Lim	Dasmarinas Cavite	09771239876	P01	Keyboard	Accessories	800	2	TechSource

Show all | Number of rows: 25 | Filter rows: Search this table

ProductID	ProductName	ProductCategories	UnitPrice	Quantities	SupplierName	SupplierContacts	PaymentMethods	SalesPersonID	SalesPersonName	SalesPersonEmail
P01, P02	Keyboard, Mouse	Accessories, Accessories	800, 500	1, 2	TechSource, TechSource	028888111, 08888111	Cash	501	Mark Reyes	mark.reyes@store.com
P03	Monitor	Display	7500	1	ScreenPro	027777222	Credit Card	502	Liza Santos	liza.santos@store.com
P02, P04	Mouse, USB Drive	Accessories, Storage	500, 600	1, 3	TechSource, DataHub	028888111, 026666333	Gcash	501	Mark Reyes	mark.reyes@store.com
P01	Keyboard	Accessories	800	2	TechSource	028888111	Gcash	503	Carlo Diaz	carlo.diaz@store.com

1NF/ 1ST NORMAL FORM

Server: 127.0.0.1 > Database: labexammidterm > Table: sales_inf

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	OrderID	int(11)			No	None			Change Drop More
☐ 2	OrderDate	date			Yes	NULL			Change Drop More
☐ 3	CustomerID	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	CustomerName	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 5	CustomerAddress	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 6	CustomerPhone	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 7	ProductID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐ 8	ProductName	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 9	ProductCategory	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 10	UnitPrice	decimal(10,2)			Yes	NULL			Change Drop More
☐ 11	Quantity	int(11)			Yes	NULL			Change Drop More
☐ 12	SupplierName	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 13	SupplierContact	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 14	PaymentMethod	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 15	SalespersonID	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 16	SalespersonName	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 17	SalespersonEmail	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

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2NF/ 2ND NORMAL FORM

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	ProductID	varchar(120)	utf8mb4_general_ci		No	None			Change Drop More
☐ 2	ProductName	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	SupplierNames	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	SupplierContacts	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	OrderID	int(11)			No	None			Change Drop More
☐ 2	OrderDate	date			Yes	NULL			Change Drop More
☐ 3	CustomerID	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	CustomerName	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 5	SalespersonID	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More

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3NF/ 3RD NORMAL FORM

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	SuppliersName	varchar(120)	utf8mb4_general_ci		No	None			Change Drop More
2	SupplierContact	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	CustomerID	varchar(120)	utf8mb4_general_ci		No	None			Change Drop More
2	CustomerName	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	CustomerAddress	varchar(120)	utf8mb4_general_ci		Yes	NULL			Change Drop More

1. Which attributes cause partial dependency?

Partial dependency exists because the table's logical primary key is composite, most likely (OrderID, ProductID), since one order can contain multiple products. Attributes that depend only on OrderID and not on the whole composite key cause partial dependency. These include OrderDate, CustomerID, PaymentMethod, and SalespersonID because they describe the order itself, not the specific product in the order. Likewise, attributes that depend only on ProductID also cause partial dependency. These include ProductNames, ProductCategories, UnitPrices, SupplierNames, and SupplierContacts, since they describe the product and not the entire order-product combination. Because these attributes depend on only part of the composite key rather than the whole key, they violate Second Normal Form (2NF).

2. Which attributes cause transitive dependency?

Transitive dependency occurs when a non-key attribute depends on another non-key attribute instead of depending directly on the primary key. In the table, CustomerName, CustomerAddress, and CustomerPhone are transitively dependent because they depend on CustomerID, which itself depends on OrderID. Similarly, SalespersonName and SalespersonEmail depend on SalespersonID, not directly on the primary key. In addition, SupplierContacts depends on SupplierNames, which is determined by ProductID, creating another transitive relationship. These attributes violate Third Normal Form (3NF) because they rely on non-key attributes rather than depending directly on the primary key.